# Assignment # 4

# DBMS

**Total Marks:** 30 **Date:** 06/06/2026 **Submission Date:** 10/06/2026 11:00am

**Name**________________ **Roll No.** ____________ **Class:** ___________

## Instructions:

I. Assignment should **be hand written,** Must be **A-4 Size Paper**.

II. Assignment should be Submit on Portal within time.

III. You should create your own solution according to your logic. Any similarity with other student and internet will be consider as cheating and your assignment will be marked as ZERO.

IV. Attach this page as front page of your assignment solution.

V. Assignment should be in Group of only 5 members.

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### Question 1:

Given a relation $R(U, V, W, X, Y, Z)$ and the following set of functional dependencies:

$$\{UV \rightarrow W, \quad W \rightarrow X, \quad X \rightarrow Y, \quad Z \rightarrow U, \quad Z \rightarrow V\}$$

Tasks:

- a) Compute the attribute closure of Z.
- b) Identify all Candidate Keys for relation R.
- c) List at least two Super Keys.
- d) Can Z be considered a Primary Key? Justify with reasoning.
- f) Explain how the dependencies above affect normalization up to 3rd.

### Question 2:

Consider a relation schema $\boldsymbol{R(A, B, C, D, E, F)}$ with the following set of functional dependencies:

$$\{\boldsymbol{A \rightarrow BC, \quad B \rightarrow D, \quad CD \rightarrow E, \quad E \rightarrow F, \quad F \rightarrow A}\}$$

Tasks:

- a) Find the closure of A and determine whether A is a candidate key.
- b) Identify all candidate keys of R.
- c) List any two super keys that are not candidate keys.
- d) Pick any one candidate key as the primary key and justify the selection.

### Question 3:

Consider the relational schema:

$\boldsymbol{R(A, B, C, D, E, F, G, H)}$ and the following functional dependencies over R:

1. $\{\boldsymbol{A \rightarrow CE, \ B \rightarrow D, \ CD \rightarrow F, \ EF \rightarrow G, \ G \rightarrow H, \ D \rightarrow B}\}$
2. $\{\boldsymbol{AB \rightarrow D, \ D \rightarrow E, \ E \rightarrow F, \ C \rightarrow G, \ G \rightarrow H, \ H \rightarrow C}\}$

identify the candidate key(s) for the above relation.

## Question 4

Consider the relational schema: $\boldsymbol{R(A,B,C,D,E)}$ and the following functional dependencies over R:

$$\boldsymbol{A \rightarrow BC}$$
$$\boldsymbol{B \rightarrow C}$$
$$\boldsymbol{A \rightarrow B}$$
$$\boldsymbol{AB \rightarrow D}$$
$$\boldsymbol{AC \rightarrow D}$$
$$\boldsymbol{D \rightarrow E}$$

Find the **minimal cover** of the given set of functional dependencies.

## Question 5

Consider the relational schema: $\boldsymbol{R(A,B,C,D,E,F,G)}$ and the following functional dependencies over R:

$$\boldsymbol{AB \rightarrow CD}$$
$$\boldsymbol{C \rightarrow E}$$
$$\boldsymbol{D \rightarrow F}$$
$$\boldsymbol{A \rightarrow C}$$
$$\boldsymbol{B \rightarrow D}$$
$$\boldsymbol{EF \rightarrow G}$$
$$\boldsymbol{ABE \rightarrow G}$$

Find the **minimal cover** of the given set of functional dependencies

## Question 6

Consider the relational schema: $\boldsymbol{R(A,B,C,D,E,H)}$ and the following functional dependencies over R:

$$\boldsymbol{A \rightarrow BC}$$
$$\boldsymbol{B \rightarrow D}$$
$$\boldsymbol{C \rightarrow D}$$
$$\boldsymbol{D \rightarrow E}$$
$$\boldsymbol{AB \rightarrow E}$$
$$\boldsymbol{E \rightarrow H}$$
$$\boldsymbol{AH \rightarrow D}$$

Find the **minimal cover** of the given set of functional dependencies.

## Question 7

Consider relation:

$$R(EmpID, ProjectID, EmpName, ProjectName, Hours, DeptID, DeptName)$$

Functional dependencies:

$$EmpID \rightarrow EmpName, DeptID$$
$$DeptID \rightarrow DeptName$$
$$ProjectID \rightarrow ProjectName$$
$$EmpID, ProjectID \rightarrow Hours$$

Identify the candidate key, prime attributes, and highest normal form.

## Question 8

Consider relation: $R(A, B, C, D, E)$

Functional dependencies: $\{A \rightarrow B,\ B \rightarrow C,\ CD \rightarrow E\}$

Identify the candidate key, prime attributes, and highest normal form.

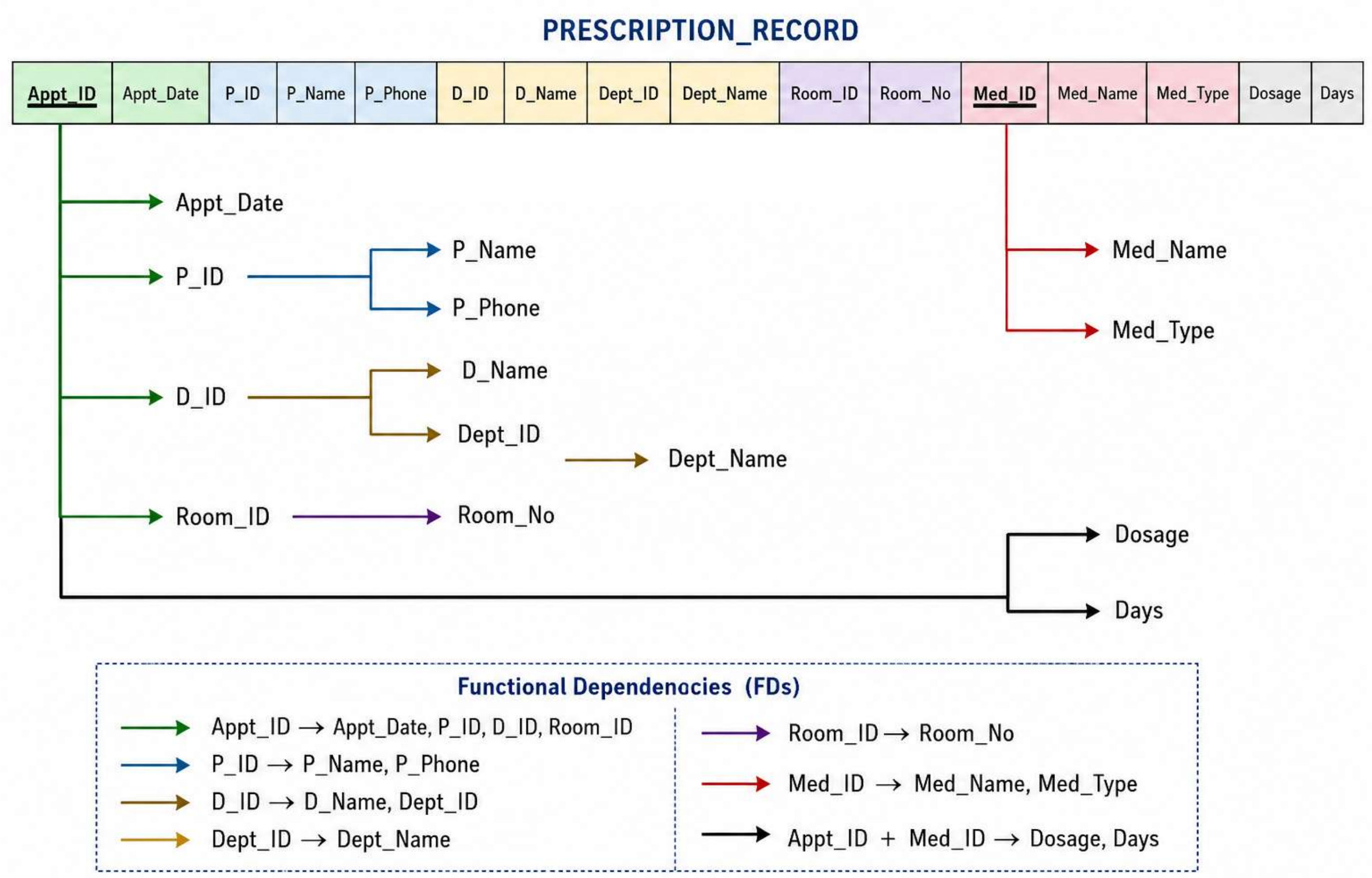
## Question 9

The given figure shows a functional dependency for a hospital prescription system. The relation is named **PRESCRIPTION_RECORD**.

The relation contains the following attributes:

`Appt_ID, Appt_Date, P_ID, P_Name, P_Phone, D_ID, D_Name, Dept_ID, Dept_Name, Room_ID, Room_No, Med_ID, Med_Name, Med_Type, Dosage, Days`

Study the given diagram carefully and answer the following questions.




**<u>Required Tasks</u>**

**a)** Identify the all-possible **candidate key** of the relation.

**b)** Identify the **prime attributes** and **non-prime attributes**.

**c)** Write all functional dependencies shown in the diagram.

**d)** Identify all **partial dependencies** in the relation.

**e)** Identify all **transitive dependencies** in the relation.

**f)** State the current highest normal form of the relation and justify your answer.

**g)** Normalize the relation into **Second Normal Form — 2NF**.

**h)** Normalize the relation into **Third Normal Form — 3NF**. Clearly mention all final tables with **primary keys** and **foreign keys**.

**Note :** You must show all normalization steps clearly. Directly writing final 3NF tables without identifying the candidate key, partial dependencies, and transitive dependencies will not receive full marks.

## Question 10

1. Produce the First, Second and Third Normal Form of this document by normalization

**Order Form**

Order number: 1234 Date: 11/04/98

Customer number: 9876

Customer name: Billy

Customer address: 456 HighTower Street

City–Country: Hong Kong, China

| ProductNo | Desscription | Quantity | Unit Price |
|---|---|---|---|
| A123 | Pencil | 100 | $3.00 |
| B234 | Eraser | 200 | $1.50 |
| C345 | Sharpener | 5 | $8.00 |

2.

**An agency called *InstantCover* supplies part-time/temporary staff to hotels throughout Scotland. The table shown in Figure 2 lists the time spent by agency staff working at two hotels. The National Insurance Number (NIN) is unique for employee.**

| NIN | contractNo | hoursPerWeek | eName | hotelNo | hotelLocation |
|---|---|---|---|---|---|
| 113567WD | C1024 | 16 | John Smith | H25 | Edinburgh |
| 234111XA | C1024 | 24 | Diane Hocine | H25 | Edinburgh |
| 712670YD | C1025 | 28 | Sarah White | H4 | Glasgow |
| 113567WD | C1025 | 16 | John Smith | H4 | Glasgow |

Figure 2: Employees of *InstantCover* and their contracts to work at hotels.

**(1) The table shown in Figure 2 is susceptible to update anomalies. Provide examples of insertion, deletion, and modification anomalies.**

2) Show the 1NF, 2NF and 3NF